CAREER GUIDANCE ACADEMY (C.G.A)

Comprehensive System Architecture & Detailed Project Overview

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ACADEMIC PLATFORM V2.0

Stack: Next.js 16 + Supabase
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Executive Summary

Career Guidance Academy (C.G.A) is an enterprise-grade coaching institute web platform and Learning Management System (LMS) specifically tailored for competitive examination aspirants and school students in Tripura. The platform seamlessly integrates a public discovery portal, an authenticated student learning environment, and a robust institute management and CMS console.

1. Project Scope & Coaching Divisions

The academy caters to four major academic categories, each with tailored curricula, mock test series, and learning materials:

1. Teaching & State Recruitment Exams

Comprehensive preparation for Tripura education and recruitment boards:

- Tripura TET Paper I & II
- STGT (Secondary Teachers)
- STPGT (Post Graduate)
- TPSC (Combined / Departmental)
- JRBT Group C & D

2. Banking, SSC & Railway Exams

National level competitive examinations training:

- IBPS PO & Clerk
- SBI PO & Junior Associates
- SSC CGL / CHSL / MTS
- RRB NTPC & Group D

3. Other Competitive Examinations

Specialized defense and higher education entrance programs:

- Tripura Police SI & Constable
- Defence Forces Recruitment
- CUET (UG Entrance)

4. School Foundation Coaching (Class VII to X)

Board-specific academic foundation coaching:

- TBSE Class VII, VIII, IX, X
- CBSE Class VII, VIII, IX, X
- Dedicated Board Batches

2. Technical Architecture & Stack

Layer	Technology	Role in Application
Frontend Framework	Next.js 16 (App Router), React 19, TypeScript 5	Server-side rendering, nested layouts, rapid initial loads, and edge optimization.
Styling & Design System	Tailwind CSS v4, Lucide Icons	Custom academic palette (Navy #0A0E2A, Royal Blue #0056B3, Torch Orange #EA580C), responsive grids, and clean cards.
Database & RLS	Supabase PostgreSQL with Row Level Security	Multi-tenant data isolation, strict authorization policies, sequences, triggers, and automated cascade scripts.
Authentication	Supabase Auth (@supabase/ssr)	Session cookies, role metadata (student vs admin), email verification, and password recovery.
Document Engine	pdf-parse & Custom RegEx Ingestion	Automated extraction of questions, options (A to D), answer keys, and explanations from uploaded PDF question papers.
State & Cache Layer	React Context (CgaContext) & Service Layer	Optimistic UI updates, in-app toast notices, reactive student session synchronization, and hybrid seed fallbacks.

3. Core Environments & Features

A. Public Web Portal (Open Access)

- **Homepage (/):** Hero banner with institute mission, 4-division coaching directory, featured courses, faculty highlights, student testimonials, and direct enquiry modal.
- **Course Catalogue (`/courses`):** Advanced search and multi-facet filtering (Category, Mode: Online/Offline, Status: Open for Admission / Batch Closed, and Board: TBSE vs CBSE).
- **Achievements (`/achievements`):** Board toppers and state rank-holders gallery filterable by exam category with scores and testimonial quotes.
- **Faculty Directory (`/faculty`):** Subject mentor profiles with designations, academic credentials, teaching experience, and specializations.
- **About Us & Contact (`/about`, `/contact`):** Campus history, director's address, embedded Udaipur Google Maps, phone, email, and validated admission inquiry form that records into the database.
- **Authentication Flows (`/login`, `/register`, `/forgot-password`):** Tabbed sign-in, self-registration generating sequential ID `CGA-YYYY-XXX`, and secure password recovery.

B. Student Learning Portal (`/student/*`)

- **Student Dashboard (`/student/dashboard`):** Welcome banner, active enrolled course cards with syllabus progress meters, scheduled upcoming tests, and fee dues status.
- **Course Self-Enrolment (`/student/courses`):** Choose between **Full Payment** or **2-Part Instalment Plan**, complete simulated checkout, auto-decrement available batch seats, download instant official fee receipt, or join waiting list when full.
- **My Courses (`/student/my-courses`):** Batch timings, assigned faculty mentor, syllabus completion tracker, and shortcuts to associated materials and tests.
- **Study Materials Library (`/student/materials`):** Gated soft-copy PDFs and paid notes unlocked only for enrolled batches. Includes an **In-Browser Anti-Piracy Watermarked Reader** that overlays the student's name, ID, and timestamp across pages.
- **Mock Tests Arena (`/student/tests`):** Timed Computer-Based Test (CBT) engine featuring:
  - Real-time countdown timer, question palette (Answered, Unanswered, Marked for Review).
  - Formula-based negative marking calculation.
  - **Exam Integrity:** Faculty solutions and answer keys remain strictly locked while taking the test.
  - Instant scorecard with batch merit rank, accuracy percentage, and full question explanations unlocked upon submission.
- **Fee Ledger & Receipts (`/student/fees`):** Comprehensive breakdown of total course fee, paid amount, pending balance, instalment clearance flow, and printable official receipts.

C. Institute Administrator & CMS Console (`/admin/*`)

- **Executive KPI Dashboard (`/admin/dashboard`):** Total student enrollment, active courses, total institute revenue, outstanding fee dues, and visual division seat-occupancy charts.
- **Student Management (`/admin/students`):** Search and filter student directory, 360° student modal (enrolled courses, fee transactions, test attempts), export roster to CSV/Excel, and generate temporary sign-in credentials.
- **Atomic Candidate Purge (Cascade Deletion):** Admin ability to permanently scrub any candidate across `test_attempts`, `fee_payments`, `enrollments`, `waiting_list`, `profiles`, and `auth.users` in a single safe atomic operation.
- **Course & Batch Control (`/admin/courses`):** Create and edit courses, adjust fee schedules, and open/close admissions in real-time.
- **Automated PDF Test Ingestion (`/admin/tests`):** Upload PDF/text question papers; intelligent parser extracts 50–100 questions, choices (A–D), answer keys, and explanations with one-click key modifications before publishing.
- **Fee Management & Defaulter Tracking (`/admin/fees`):** Record counter cash/UPI payments, issue receipts, and filter fee defaulters by batch.

- **Content Management System:** Dedicated management interfaces for Study Materials (`/admin/materials`), Faculty (`/admin/faculty`), Achievements (`/admin/achievements`), Announcements (`/admin/announcements`), and Live Site Content (`/admin/site-content`).

4. Database Schema & Relational Design

Table Name	Keys & Relationships	Core Attributes & Responsibilities
`profiles`	`id` (UUID PK → auth.users)	role ('student'|'admin'), student_code (Unique 'CGA-2026-XXX'), name, email, phone, guardian_name, qualification, target_exam, status ('Active'|'Inactive').
`courses`	`id` (Text PK), faculty_id (FK)	name, category, mode ('Online'|'Offline'), board (TBSE/CBSE), class_level, price, first_instalment, batch_name, batch_size, seats_available, waiting_list_count, status ('Open for Admission'|'Batch Closed').
`faculty`	`id` (Text PK)	name, designation, subject, qualification, experience, photo, bio, specialization_tags.
`enrollments`	`id` (UUID PK), student_id, course_id	enrolled_date, progress (0–100%), total_fee, paid_amount, due_amount, next_due_date. Unique constraint on (student_id, course_id).
`fee_payments`	`id` (UUID PK), receipt_no (UK)	student_id, course_id, amount, payment_mode (UPI, Card, Net Banking, Cash), payment_type (Full Payment, Instalment 1/2, Balance Clearance), date, remarks.
`mock_tests`	`id` (Text PK), course_id (FK)	title, batch_name, duration_minutes, total_marks, marks_per_question, negative_marking, status ('Published'|'Draft'), questions (JSONB array with answer keys & explanations).
`test_attempts`	`id` (UUID PK), test_id, student_id	score, total_marks, correct_count, wrong_count, unattempted_count, accuracy, time_taken_minutes, submitted_at, answers (JSONB submitted choices).
`study_materials`	`id` (Text PK), course_id (FK)	title, type ('Book PDF'|'Paid Notes'), author, pages, file_size, preview_sections (JSONB), uploaded_date.
`announcements`	`id` (Text PK)	title, content, target_batch ('All Batches' or specific batch), priority ('Normal'|'Important'|'Urgent'), date, author.
`enquiries`	`id` (UUID PK)	name, phone, email, course_interest, message, created_at.
`waiting_list`	(student_id, course_id) Composite PK	Auto-queued students when course seats are exhausted.
`site_content`	`id` (Int PK = 1)	CMS-managed JSONB storing hero banners, institute mission/vision, and contact phone/address.

5. Security, Row Level Security (RLS) & RPCs

Row Level Security (RLS) Policies

- **Public:** Read-only access to published courses, faculty, achievements, and CMS site content. Can insert public inquiries.
- **Students:** Strictly restricted to their own profiles, their own enrollment records, unlocked course materials, fee ledger, and test scores.
- **Admins:** Evaluated through Postgres role checks and authenticated via the Supabase Service Role Key on backend server actions.

Transactional Stored Procedures (RPCs)

- `submit_mock_test_attempt()`: Server-authoritative calculation of test scores, negative markings, accuracy, and attempt records.
- `enrol_student()`: Atomically records enrollment, creates payment receipt, and decrements batch seats within a single transaction.
- `purge_candidate()`: Admin cascade that purges all student records across five tables plus authentication credentials.

6. Unique Architectural Innovations

1. Bulk AI / RegEx Question Ingestion

Parses complex question formats: numbered questions (`1.`, `Q1.`), option choices (`(A)-(D)`, `A)-D)`), inline answer keys (`Ans: B`), and separate bottom answer tables. Eliminates hours of manual data entry.

2. Dynamic Document Watermarking

Watermarks PDF study notes in real-time with the active student's name, roll code, and timestamp, effectively preventing study material piracy.